

Trend of EV Adoption in Singapore

Project Proposal

1. Aims, Objectives and Background

1.1 Introduction

In this ever-changing landscape of global transportation, the paradigm is gearing towards sustainable alternatives, and being one of the fastest developed countries in the world, Singapore is at the forefront of this vehicular revolution. The increasing concerns about environmental sustainability has fueled the adoption of electric vehicles (EVs) in this tiny country and worldwide.

This project is an attempt to dig deeper into the factors affecting the trend of electric vehicle adoption in Singapore, observe the trend and challenges, and if possible, predict the future prospects of EVs in Singapore.

1.2 Motivation and Background

The Singapore government, specifically the Land Transport Authority (LTA), has decided on an ambitious goal to reduce land transport emissions by 80% by the mid-century.[1] Given Singapore's reliance on natural gas for power generation, transitioning from internal combustion engine (ICE) vehicles to electric vehicles (EVs) is a sustainable choice. With this in mind, they've introduced the "Singapore Green Plan 2030", a comprehensive EV Roadmap to drive the adoption of electric vehicles. Under this roadmap, Singapore aims to establish 60,000 EV charging points by 2030: 40,000 in public carparks and 20,000 in private premises. One important thing to note is that, the public housing board in Singapore i.e., Housing and Development Board (HDB) endeavors each HDB town to be an EV-Ready Town, with around 2,000 carparks equipped with charging points by 2025.

However, every Singaporean knows how expensive owning a car is in the country right now with the increasing COE (Certificate of Entitlement) prices; the next thing that's more expensive is owning an apartment in Singapore, even one under HDB.

Will the hoi polloi of the nation be able to adopt this new trend of transitioning to a much more expensive, albeit sustainable, way of owning a car? Or will the charging points installed ultimately be a waste of time, effort and money by the government? That's the question this project is aiming to explore.

1.3 Aims and Objectives

This report aims to use data science webscraping to investigate the trend of EV adoption currently (i.e., volume of new registration of EV cars by make, EV population), the reasons for the adoption/non-adoption and the general sentiments the locals have towards electric

vehicles. A further analysis might be taken to find out the demographics of people buying EVs within the data period specified in the project scope. The next step will be to source for another country's EV adoption trend (i.e., Norway) to compare the findings.

1.4 Scope

Having spoken to a personnel working at a local private company involved in the installation of EV charging stations in Singapore, I've decided that the scope of this project will deal with data and information within the last 3 years (2021 - 2023) as Singapore is still relatively new to the EV scene. Also, from the dataset, I will only be looking at electric vehicles, hybrids like petrol-electric or diesel-electric will not be considered in order to align with the goals of the EV Roadmap. Taxis and other form of electric vehicles like motorcycles and buses will not be taken into consideration as well for the purpose of this project.

2. Data

2.1 Data Requirements and Chosen Organisations

I will be using statistical data and publications from the LTA website (including its subsidiary OneMotoring), local newspaper website The Straits Times and also the local English news channel - Channel News Asia (CNA)'s website to webscrape for information for this project.

2.2 Type of Data

2 sets of static csv files will be used:

- Monthly New Registration of Cars by Make (DataMall, LTA)
- Monthly Motor Vehicle Population Statistics by Type of Fuel Used (DataMal, LTA)

3. Limitations

3.1 Limitations of Approach

One limitation is that the qualitative data gathered from the LTA, The Straits Times and CNA websites might be skewed as those three are government websites. Extreme or negative sentiments might be filtered. As EVs are still relatively new to the country, there are not enough forums or publications at the moment to paint a more accurate picture of the EV adoption scene in the country.

3.2 Limitations of dataset

The dataset is only exploring a period of 3 years, this may or may not be a long enough period to show a significant trend.

For CSV file - Monthly New Registration of Cars by Make, as EVs have only been appearing in Singapore during the recent years, they were documented as "Others" under "fuel_type", it's only in the current times that they are labelled as "electric" in the dataset. "Others" can also mean non-EV type of cars as well, so that may affect the accuracy of the analysis. Nonetheless, for this project's purpose, we'll include "Others" numbers in our analysis as well.

4. Ethics

4.1 Ownership and Provenance

The data sourced from the Land Transport Authority (LTA) website, local newspaper website The Straits Times and Channel News Asia (CNA)'s website are handled with transparency regarding its provenance. Acknowledgment of the ownership of the data, highlighting that it comes from the LTA and CNA is mandatory in this project.

4.2 Data Preparation, Refinement, and Exploration

The two sets of CSV files obtained from LTA are used in accordance with the terms of use specified by LTA. Any insights or analysis derived from the data will be attributed to the original sources.

4.3 Potential Biases in the Dataset

As these data are retrieved from government websites, spotting inherent biases that might exist is crucial and will be mitigated during analysis.

5. Libraries and Modules

```
In [1]: ▶ # Import Libraries and Modules
import pandas as pd
from tabulate import tabulate
from decimal import Decimal
from bs4 import BeautifulSoup
import json
import requests
import nltk
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from wordcloud import WordCloud, ImageColorGenerator
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
import numpy as np
from PIL import Image
import re
from re import sub
import time
from nltk.stem import WordNetLemmatizer

# Set all graphs to a seaborn style with a grey background grid which matches the notebook theme
# sns.set()
```

6. Extraction of Data and Webscraping

6.1 Extract data from CSV File

6.1.1 Monthly Motor Vehicle Population Statistics by Type of Fuel Used

Let's first clean the data from the 1st CSV file.

```
In [22]: ▶ df = pd.read_csv('M09-Vehs_by_Fuel_Type.csv', encoding = "utf-8")
```

A. Clean data to filter out only electric cars for years 2021 to 2023

```
In [55]: # Convert the 'month' column to a datetime object for easy comparison
df['month'] = pd.to_datetime(df['month'])

# Filter for the years 2021 to 2023, 'Electric' type, and 'Cars' category
filtered_df = df[(df['month'].dt.year >= 2021) &
                  (df['month'].dt.year <= 2023) &
                  (df['type'] == 'Electric') &
                  (df['category'] == 'Cars')]

# Convert the 'number' column to integers
filtered_df.loc[:, 'number'] = filtered_df['number'].astype(int)

# Display the filtered dataframe
print(tabulate(filtered_df, headers='keys', tablefmt='pretty'))
```

	month	category	type	number
1709	2021-01-01 00:00:00	Cars	Electric	1274
1738	2021-02-01 00:00:00	Cars	Electric	1307
1767	2021-03-01 00:00:00	Cars	Electric	1351
1796	2021-04-01 00:00:00	Cars	Electric	1415
1825	2021-05-01 00:00:00	Cars	Electric	1485
1854	2021-06-01 00:00:00	Cars	Electric	1549
1883	2021-07-01 00:00:00	Cars	Electric	1643
1912	2021-08-01 00:00:00	Cars	Electric	1855
1941	2021-09-01 00:00:00	Cars	Electric	2243
1970	2021-10-01 00:00:00	Cars	Electric	2309
1999	2021-11-01 00:00:00	Cars	Electric	2635
2028	2021-12-01 00:00:00	Cars	Electric	2942
2057	2022-01-01 00:00:00	Cars	Electric	3149
2086	2022-02-01 00:00:00	Cars	Electric	3297
2115	2022-03-01 00:00:00	Cars	Electric	3576
2144	2022-04-01 00:00:00	Cars	Electric	3764
2173	2022-05-01 00:00:00	Cars	Electric	4054
2202	2022-06-01 00:00:00	Cars	Electric	4421
2231	2022-07-01 00:00:00	Cars	Electric	4716
2260	2022-08-01 00:00:00	Cars	Electric	4990
2289	2022-09-01 00:00:00	Cars	Electric	5421
2318	2022-10-01 00:00:00	Cars	Electric	5779
2348	2022-11-01 00:00:00	Cars	Electric	6124
2378	2022-12-01 00:00:00	Cars	Electric	6531
2408	2023-01-01 00:00:00	Cars	Electric	6809
2438	2023-02-01 00:00:00	Cars	Electric	7048
2468	2023-03-01 00:00:00	Cars	Electric	7314
2498	2023-04-01 00:00:00	Cars	Electric	7615
2528	2023-05-01 00:00:00	Cars	Electric	7961
2558	2023-06-01 00:00:00	Cars	Electric	8390
2588	2023-07-01 00:00:00	Cars	Electric	8873
2618	2023-08-01 00:00:00	Cars	Electric	9454
2648	2023-09-01 00:00:00	Cars	Electric	9998
2678	2023-10-01 00:00:00	Cars	Electric	10426
2708	2023-11-01 00:00:00	Cars	Electric	10983

B.Graphical Representation: Plot line graphs to show trend across each year

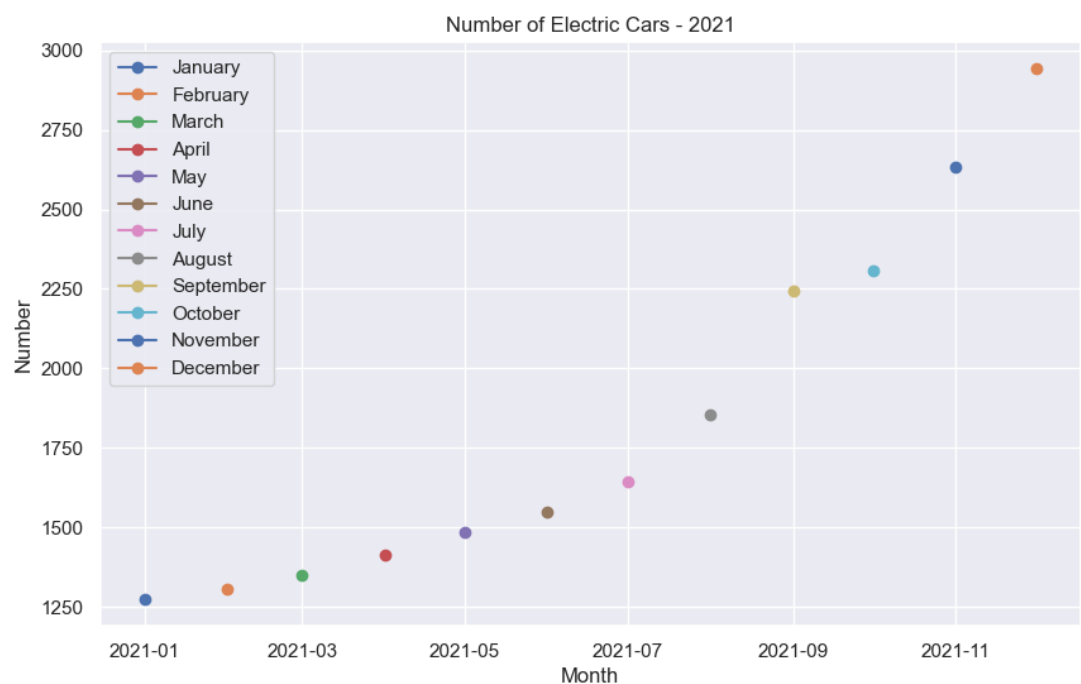
```
In [47]: # Convert the 'month' column to a datetime object for easy comparison
df['month'] = pd.to_datetime(df['month'])

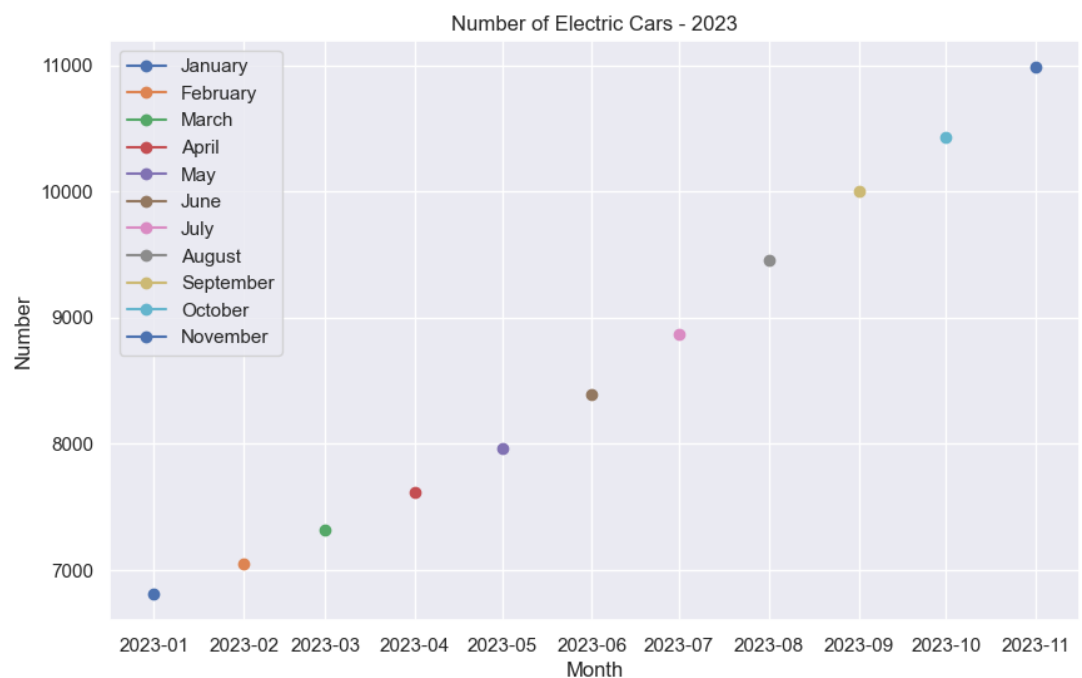
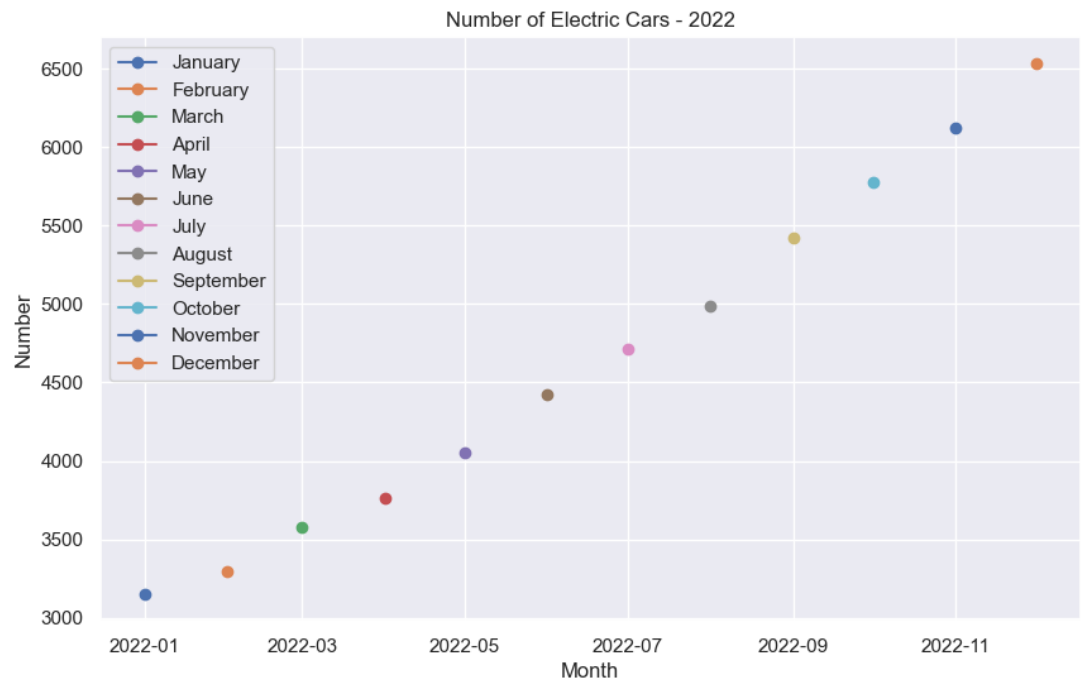
# Filter for the years 2021 to 2023, 'Electric' type, and 'Cars' category
filtered_df = df[(df['month'].dt.year >= 2021) &
                 (df['month'].dt.year <= 2023) &
                 (df['type'] == 'Electric') &
                 (df['category'] == 'Cars')]

# Convert the 'number' column to integers
filtered_df.loc[:, 'number'] = filtered_df['number'].astype(int)

# Create a separate line graph for each year
for year, group in filtered_df.groupby(filtered_df['month'].dt.year):
    plt.figure(figsize=(10, 6))
    for index, row in group.iterrows():
        plt.plot(row['month'], row['number'], marker='o', label=f"{row['month']}")

    plt.title(f'Number of Electric Cars - {year}')
    plt.xlabel('Month')
    plt.ylabel('Number')
    plt.legend()
    plt.show()
```





Analysis: We can tell from graph for 2021 that there was an overall increasing trend over the months during that year. This has sustained till 2022 but as we reached 2023 the trend has dipped slightly. We can corroborate this trend with the sentiment analysis from webscraping the chosen articles from the news websites to understand the factors causing the trend.

C. Graphical representation: Plot line graphs to compare the 3 annual trends together in one graph

```
In [56]: # Convert the 'month' column to a datetime object for easy comparison
df['month'] = pd.to_datetime(df['month'])

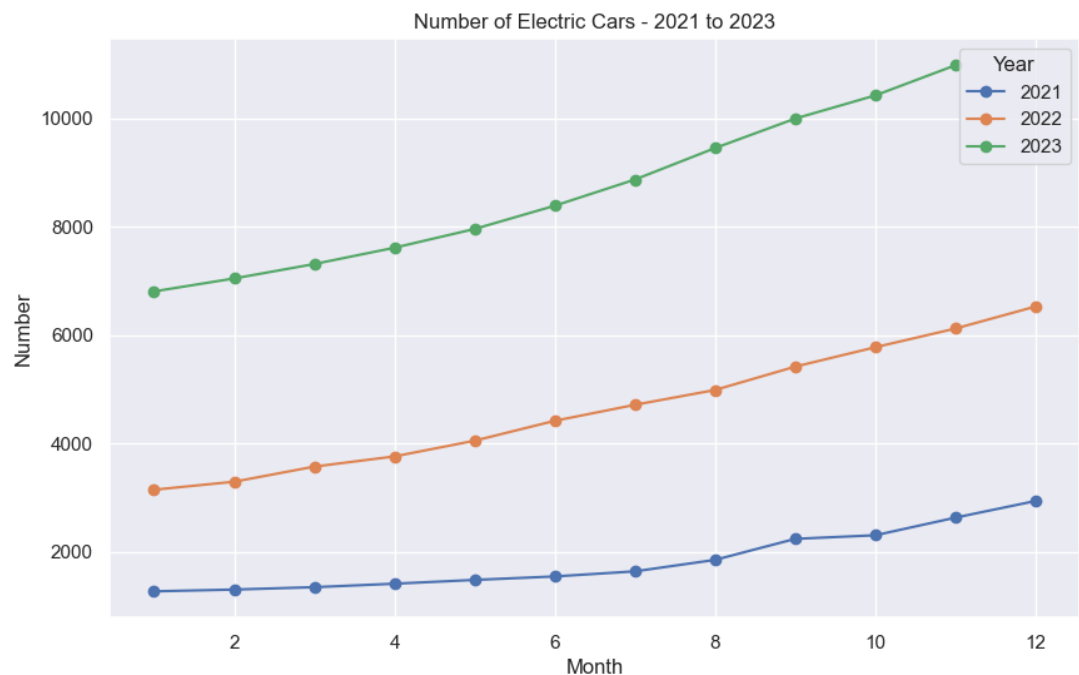
# Filter for the years 2021 to 2023, 'Electric' type, and 'Cars' category
filtered_df = df[(df['month'].dt.year >= 2021) &
                 (df['month'].dt.year <= 2023) &
                 (df['type'] == 'Electric') &
                 (df['category'] == 'Cars')]

# Convert the 'number' column to integers
filtered_df.loc[:, 'number'] = filtered_df['number'].astype(int)

# Create a single line graph with different colors for each year
plt.figure(figsize=(10, 6))

for year, group in filtered_df.groupby(filtered_df['month'].dt.year):
    plt.plot(group['month'].dt.month, group['number'], label=f"{year}")

plt.title('Number of Electric Cars - 2021 to 2023')
plt.xlabel('Month')
plt.ylabel('Number')
plt.legend(title='Year')
plt.show()
```



Analysis: This graph here shows a distinct jump in the EV population between 2022 and 2023. Again, we can find more information through sentiment analysis of the news articles to find out what caused this big jump.

D. Final population of electric cars for each year


```
In [82]: ▶ # Convert 'number' column to Decimal and then calculate the sum for each
sum_by_year = filtered_df.groupby(filtered_df['month'].dt.year)['number'

# Rename the index to 'Year'
sum_by_year.index.name = 'Year'

# Create a DataFrame from the Series
sum_by_year_df = pd.DataFrame(sum_by_year).reset_index()

# Display the sum for each year using tabulate
print(tabulate(sum_by_year_df, headers='keys', tablefmt='pretty'))
```

```
+---+-----+-----+
|   | Year | number |
+---+-----+-----+
| 0 | 2021 | 15294  |
| 1 | 2022 | 10071  |
| 2 | 2023 | 4506   |
+---+-----+-----+
```

6.1.2 Monthly New Registration of Cars by Make

Let's first clean the data from the 2nd CSV file.

A. Clean data to filter out only electric cars for years 2021 to 2023

```
In [76]: df = pd.read_csv('M03-Car_Regn_by_make.csv', encoding='utf-8')

# Convert the 'month' column to a datetime object for easy comparison
df['month'] = pd.to_datetime(df['month'])

# Filter for the years 2021 to 2023, 'Electric' type, and 'Cars' category
filtered_df = df[(df['month'].dt.year >= 2021) &
                  (df['month'].dt.year <= 2023) &
                  (df['fuel_type'].isin(['Electric', 'Others']))]

# Drop rows with non-finite values in the 'number' column
filtered_df = filtered_df.dropna(subset=['number'])

# Convert the 'number' column to integers
filtered_df['number'] = filtered_df['number'].astype(int)

# Remove the 'importer_type' column
filtered_df = filtered_df.drop(columns=['importer_type'])

# Remove the 'vehicle_type' column
filtered_df = filtered_df.drop(columns=['vehicle_type'])

# Display the first 5 and last 5 rows with ellipses in between
print(tabulate(pd.concat([filtered_df.head(), pd.DataFrame(index=['...'])]),
               headers=['keys'], tablefmt='pretty'))

# Convert 'number' column to Decimal and then calculate the sum for each year
sum_by_year = filtered_df.groupby(filtered_df['month'].dt.year)['number'].sum()

# Rename the index to 'Year'
sum_by_year.index.name = 'Year'

# Create a DataFrame from the Series
sum_by_year_df = pd.DataFrame(sum_by_year).reset_index()

# Display the sum for each year using tabulate
print(tabulate(sum_by_year_df, headers='keys', tablefmt='pretty'))
```

	month	make	fuel_type	number
20939	2021-01-01 00:00:00	AUDI	Others	33.0
20943	2021-01-01 00:00:00	BYD	Others	15.0
20948	2021-01-01 00:00:00	HONDA	Others	7.0
20950	2021-01-01 00:00:00	HYUNDAI	Others	7.0
20969	2021-01-01 00:00:00	MAZDA	Others	36.0
...	NaT	nan	nan	nan
39375	2023-11-01 00:00:00	TESLA	Electric	28.0
39379	2023-11-01 00:00:00	TOYOTA	Electric	2.0
39389	2023-11-01 00:00:00	VOLVO	Electric	9.0
39424	2023-11-01 00:00:00	FIAT	Electric	1.0
39495	2023-11-01 00:00:00	SKODA	Electric	1.0

	Year	number
0	2021	15294
1	2022	10071
2	2023	4506

B. Top makes for each year

In [80]:

```
# Convert the 'month' column to a datetime object for easy comparison
df['month'] = pd.to_datetime(df['month'])

# Filter for the years 2021 to 2023, 'Electric' type, and 'Cars' category
filtered_df = df[(df['month'].dt.year.isin([2021, 2022, 2023])) &
                  (df['fuel_type'].isin(['Electric', 'Others']))]

# Drop rows with non-finite values in the 'number' column
filtered_df = filtered_df.dropna(subset=['number'])

# Convert the 'number' column to integers
filtered_df['number'] = filtered_df['number'].astype(int)

# Group by 'make', 'fuel_type', and 'year', sum the 'number' column
make_sum = filtered_df.groupby(['make', 'fuel_type', filtered_df['month'].dt.year])

# Sort the values in descending order
make_sum = make_sum.sort_values(ascending=False)

# Create tables and plot graphs for each year
for year, group in make_sum.groupby(level=2):
    make_table = pd.DataFrame({'Make': group.index.get_level_values('make'),
                              'Fuel Type': group.index.get_level_values('fuel_type'),
                              'Total Number': group.values})

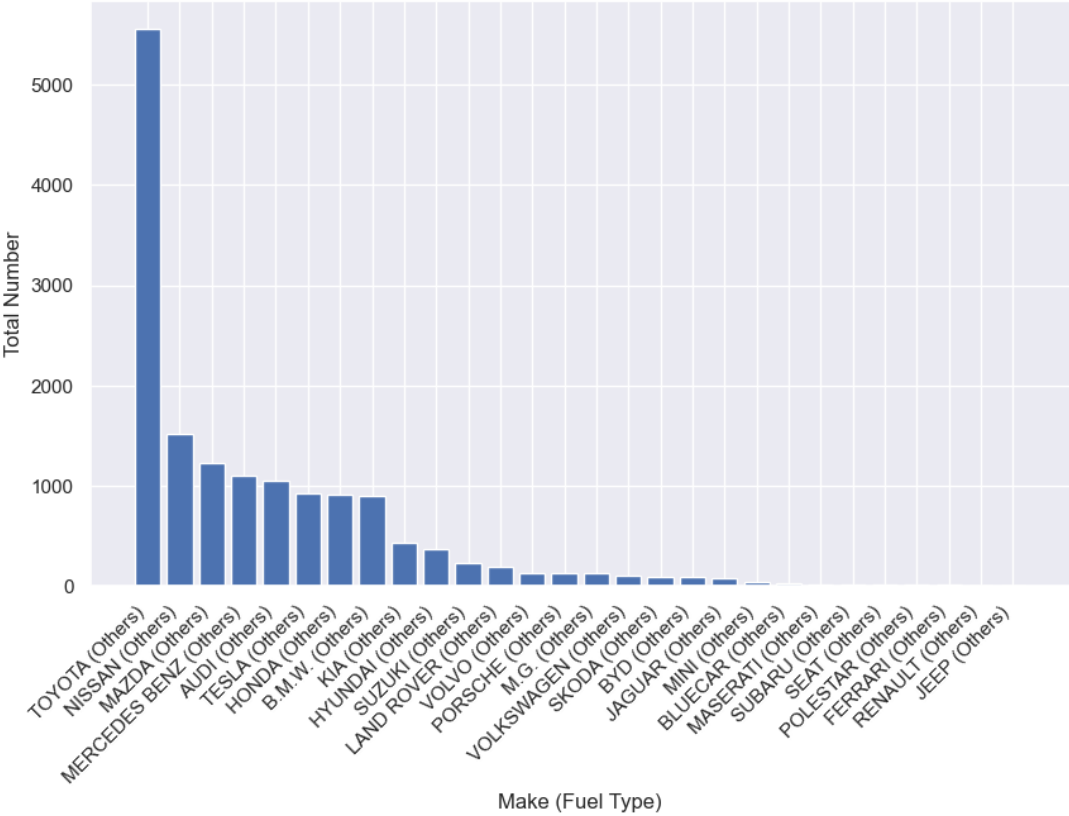
    # Display the table
    print(f"\nTop Makes for the Year {year}:\n")
    print(tabulate(make_table, headers='keys', tablefmt='pretty'))

    # Bar chart for better visualization
    plt.figure(figsize=(10, 6))
    plt.bar(make_table['Make'] + ' (' + make_table['Fuel Type'] + ')', make_table['Total Number'])
    plt.xlabel('Make (Fuel Type)')
    plt.ylabel('Total Number')
    plt.title(f'Top Makes for the Year {year}')
    plt.xticks(rotation=45, ha='right')
    plt.show()
```

Top Makes for the Year 2021:

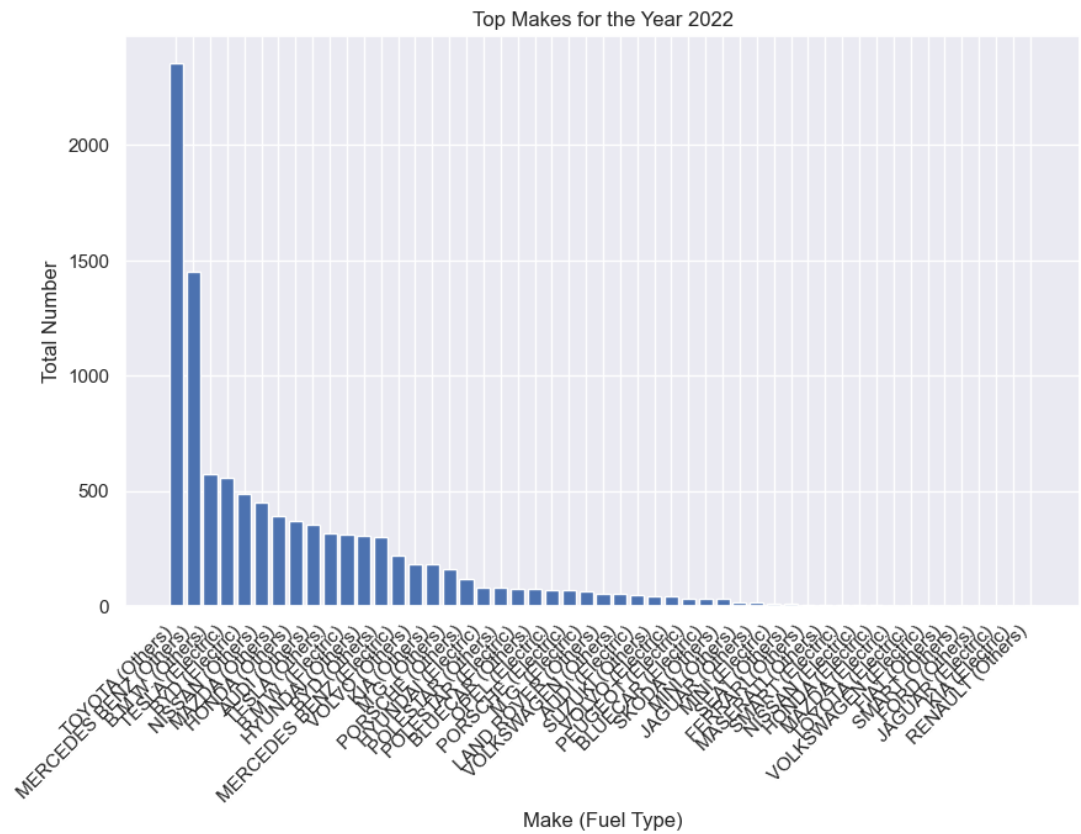
	Make	Fuel Type	Total Number
0	TOYOTA	Others	5557
1	NISSAN	Others	1519
2	MAZDA	Others	1228
3	MERCEDES BENZ	Others	1103
4	AUDI	Others	1056
5	TESLA	Others	924
6	HONDA	Others	905
7	B.M.W.	Others	900
8	KIA	Others	428
9	HYUNDAI	Others	372
10	SUZUKI	Others	224
11	LAND ROVER	Others	189
12	VOLVO	Others	134
13	PORSCHE	Others	131
14	M.G.	Others	129
15	VOLKSWAGEN	Others	105
16	SKODA	Others	93
17	BYD	Others	89
18	JAGUAR	Others	81
19	MINI	Others	39
20	BLUECAR	Others	30
21	MASERATI	Others	12
22	SUBARU	Others	11
23	SEAT	Others	11
24	POLESTAR	Others	10
25	FERRARI	Others	9
26	RENAULT	Others	4
27	JEEP	Others	1

Top Makes for the Year 2021



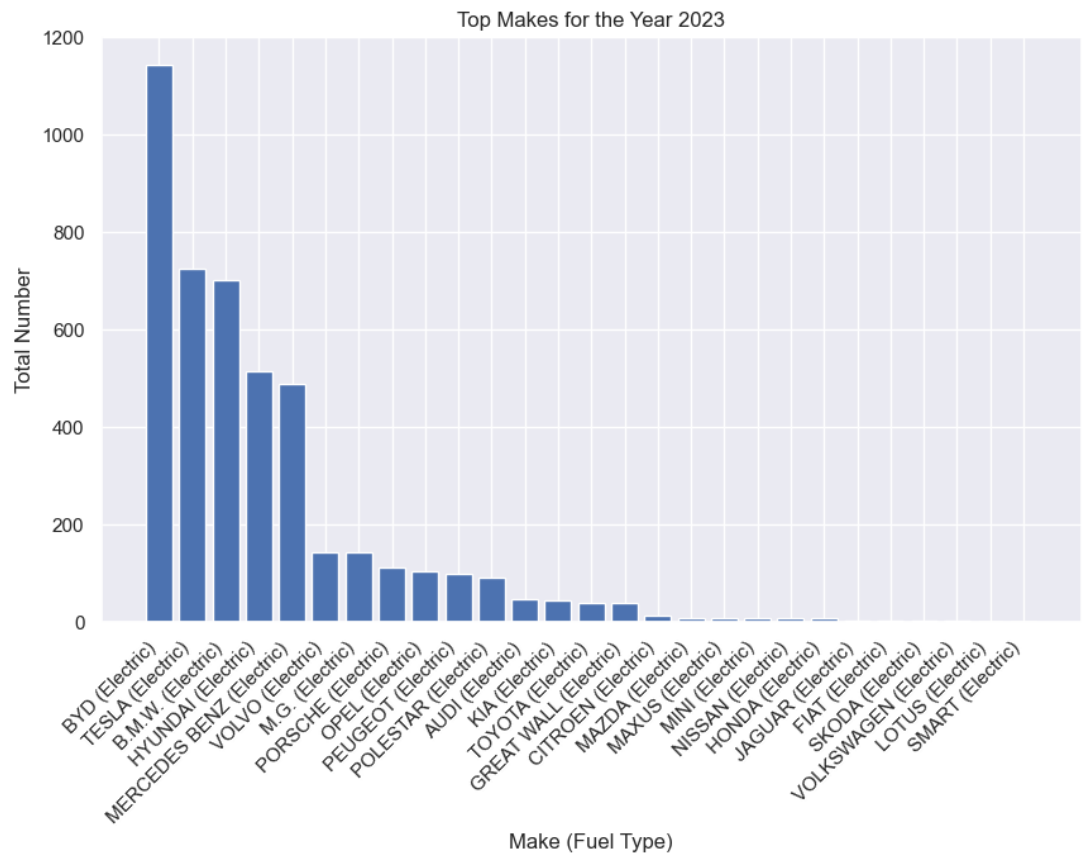
Top Makes for the Year 2022:

	Make	Fuel Type	Total Number
0	TOYOTA	Others	2354
1	MERCEDES BENZ	Others	1453
2	B.M.W.	Others	575
3	TESLA	Electric	560
4	BYD	Electric	486
5	NISSAN	Others	450
6	MAZDA	Others	389
7	HONDA	Others	370
8	AUDI	Others	354
9	TESLA	Others	315
10	B.M.W.	Electric	314
11	HYUNDAI	Others	307
12	BYD	Others	300
13	MERCEDES BENZ	Electric	219
14	VOLVO	Others	183
15	KIA	Others	182
16	M.G.	Others	163
17	PORSCHE	Others	118
18	HYUNDAI	Electric	82
19	POLESTAR	Others	81
20	POLESTAR	Electric	77
21	BLUECAR	Others	75
22	OPEL	Electric	73
23	PORSCHE	Electric	71
24	M.G.	Electric	67
25	LAND ROVER	Others	56
26	VOLKSWAGEN	Others	55
27	AUDI	Electric	49
28	SUZUKI	Others	46
29	VOLVO	Electric	43
30	PEUGEOT	Electric	34
31	BLUECAR	Electric	34
32	SKODA	Others	32
33	MINI	Others	19
34	JAGUAR	Others	17
35	MINI	Electric	14
36	SEAT	Others	11
37	FERRARI	Others	8
38	MASERATI	Others	8
39	SMART	Electric	5
40	NISSAN	Electric	5
41	HONDA	Electric	4
42	MAZDA	Electric	3
43	TOYOTA	Electric	2
44	VOLKSWAGEN	Electric	2
45	FIAT	Others	1
46	SMART	Others	1
47	FORD	Others	1
48	JAGUAR	Electric	1
49	KIA	Electric	1
50	RENAULT	Others	1



Top Makes for the Year 2023:

	Make	Fuel Type	Total Number
0	BYD	Electric	1143
1	TESLA	Electric	725
2	B.M.W.	Electric	701
3	HYUNDAI	Electric	514
4	MERCEDES BENZ	Electric	488
5	VOLVO	Electric	143
6	M.G.	Electric	143
7	PORSCHE	Electric	111
8	OPEL	Electric	105
9	PEUGEOT	Electric	100
10	POLESTAR	Electric	92
11	AUDI	Electric	47
12	KIA	Electric	44
13	TOYOTA	Electric	39
14	GREAT WALL	Electric	39
15	CITROEN	Electric	14
16	MAZDA	Electric	9
17	MAXUS	Electric	9
18	MINI	Electric	9
19	NISSAN	Electric	9
20	HONDA	Electric	8
21	JAGUAR	Electric	4
22	FIAT	Electric	4
23	SKODA	Electric	2
24	VOLKSWAGEN	Electric	2
25	LOTUS	Electric	1
26	SMART	Electric	1



6.2 Webscraping from websites

6.2.1 Defining functions for Web Scrapping

```
In [90]: ▶ def get_soup(url, jar=None):
# Create a session for better cookie handling
session = requests.Session()

# Check page accessibility
try:
    if jar:
        response = session.get(url, cookies=jar)
    else:
        response = session.get(url)
        response.raise_for_status()
except requests.exceptions.RequestException as e:
    print(f"Error accessing {url}: {e}")
    return None, None

print(response.url)
data = response.text
soup = BeautifulSoup(data, 'html.parser')
return soup, session.cookies
```

```
In [91]: soup, jar = get_soup('https://www.channelnewsasia.com/singapore/charging-jar
```



<https://www.channelnewsasia.com/singapore/charging-points-importing-rules-hamper-electric-vehicle-adoption-singapore-3408101> (<https://www.channelnewsasia.com/singapore/charging-points-importing-rules-hamper-electric-vehicle-adoption-singapore-3408101>)

```
Out[91]: <RequestsCookieJar[]>
```

6.2.2 Singaporeans' attitude towards EV adoption

A. Wordcloud to spot recurring words

```
In [118]: # Download NLTK stopwords data
nltk.download('stopwords')

def get_article_info(url):
    response = requests.get(url)
    response.raise_for_status()

    soup = BeautifulSoup(response.content, 'html.parser')

    # Extract text content from the article
    content_elements = soup.find_all('div', class_='content')
    content = ' '.join(element.get_text() for element in content_elements)

    return {'content': content.strip()}

url = 'https://www.channelnewsasia.com/singapore/charging-points-importi
article_info = get_article_info(url)

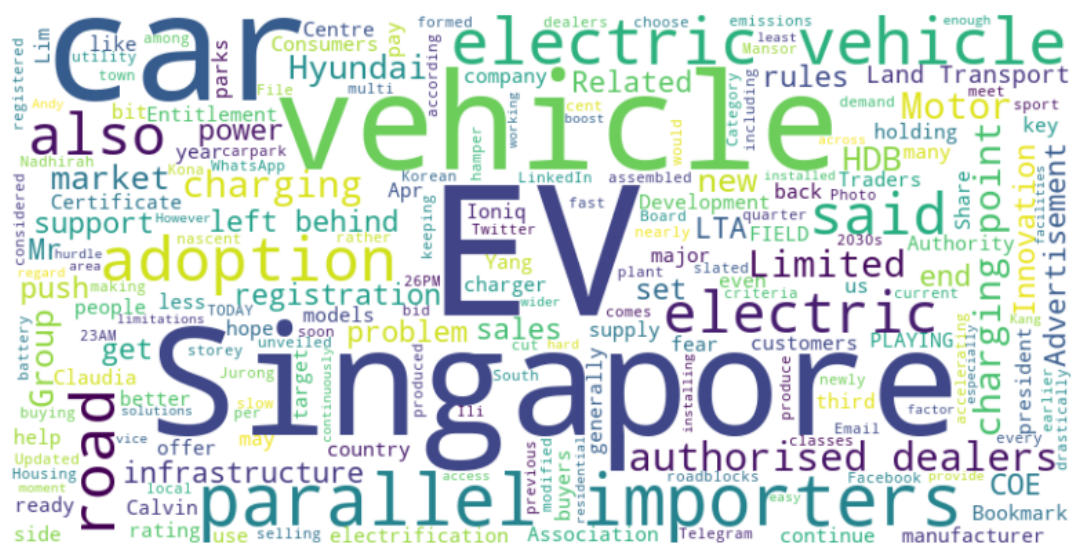
# Generate Word Cloud excluding common English stopwords
if article_info and 'content' in article_info and article_info['content']:
    content = article_info['content']

    # Use NLTK stopwords in addition to the default set
    stopwords_set = set(stopwords.words('english'))

    # Create WordCloud with stopwords
    wordcloud = WordCloud(width=800, height=400, background_color='white')

    # Plot the WordCloud image
    plt.figure(figsize=(10, 5))
    plt.imshow(wordcloud, interpolation='bilinear')
    plt.axis('off')
    plt.show()
else:
    print("No valid content found for word cloud.")
```

```
[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\devon\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!
```



7. Conclusion

The data analysis from the 2 CSV files obtained from LTA shows an increasing trend in electric vehicle population from 2021 to 2023, which shows a positive adoption of EVs in Singapore. Additionally, we discovered that BYD and Tesla are gaining more popularity as well across the 3 years, which can be explained by the two companies' increasing R&D effort at releasing new and more efficient models of cars to the EV market.

8. References

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